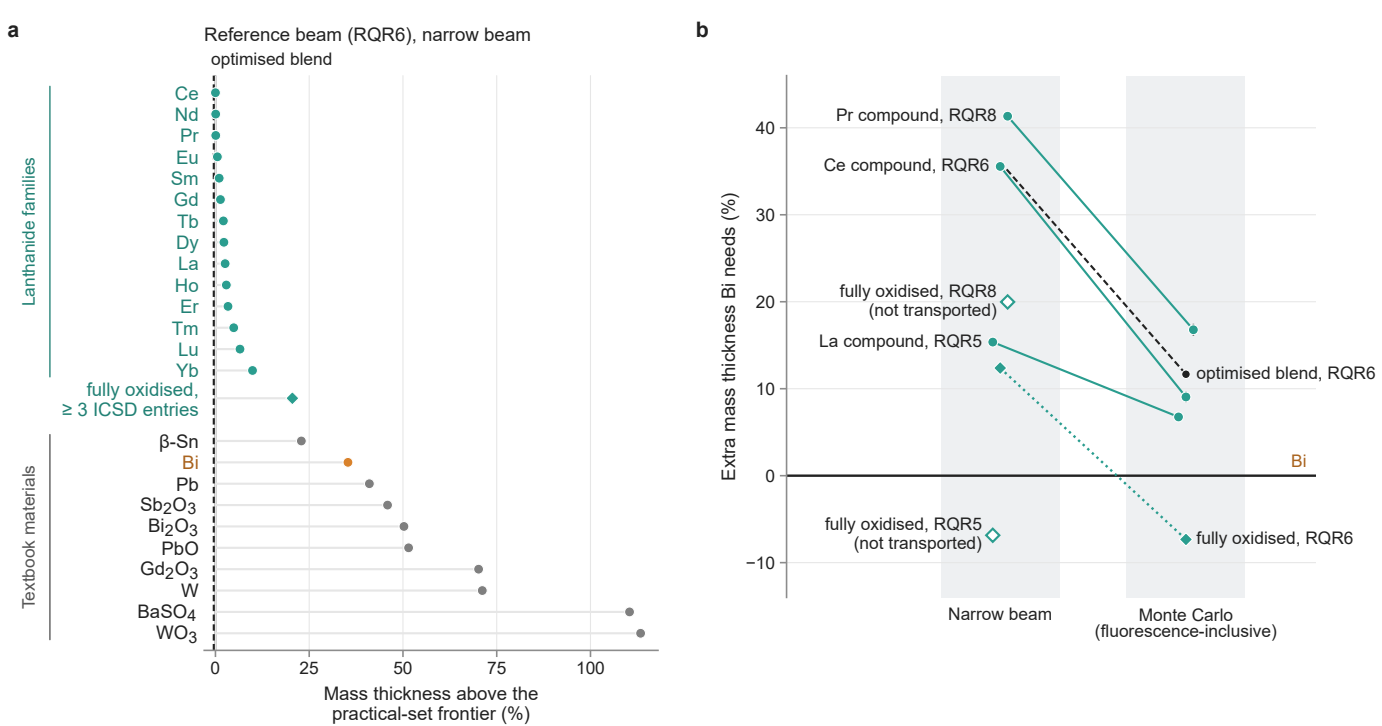
a
Reference beam (RQR6), narrow beam
optimised blend
Lanthanide families
Textbook materials
Ce
Nd
Pr
Eu
Sm
Gd
Tb
Dy
La
Ho
Er
Tm
Lu
Yb
fully oxidised,
≥ 3 ICSD entries
β-Sn
Bi
Pb
Sb2O3
Bi2O3
PbO
Gd2O3
W
BaSO4
WO3
0
25
50
75
100
Mass thickness above the
practical-set frontier (%)
b
Extra mass thickness Bi needs (%)
40
30
20
10
0
−10
Pr compound, RQR8
Ce compound, RQR6
fully oxidised, RQR8
(not transported)
La compound, RQR5
optimised blend, RQR6
Bi
fully oxidised, RQR5
(not transported)
fully oxidised, RQR6
Narrow beam
Monte Carlo
(fluorescence-inclusive)